Taming the Bullwhip Effect: A Review of AI and Multi-Agent Systems for Resilient Supply Chain Management

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***Abstract*—The level of volatility apparent in today's supply chains is on the rise, and the Bullwhip effect (fluctuating demand leads to variability upstream) is one of the largest concerns for efficiency, especially for Micro, Small and Medium Enterprises (MSMEs). In light of these issues, resilience has been increasingly spoken about, and there is a renewed interest in decentralized structures using Artificial Intelligence (AI) approaches. This literature review has built a discussion of recent academic papers to review AI and Multi-Agent Systems (MAS) and their potential to address the Bullwhip effect and increase resilience in the supply chain. The call for this review is because MSMEs represent an extraordinarily important percentage of economies and typically do not have AI options or experts, which are both accessible and available. The content presented in the themes will be as follows: decentralized decision-making systems, advanced forecasting (to include Long Short-term Memory, LSTM, networks), and new trends in AI (to include Monte Carlo Tree Search, MCTS)-like options for inventory management. A significant trend is moving from siloed processes to integrated, to autonomous, forecasting and inventory decisions jointly optimized. To summarize, we propose an AI-Native Multi-Agent architecture with custom agents tasked with coordinating the data ingestion, forecasting, and optimization activities under a common AI decision-making process that is easily understood by a non-technical user. The primary implication is the democratization of the use of advanced AI operational stability and competitive growth is realized by MSMEs. Future implications would involve extending the architectures to incorporate sustainability and circular economy perspectives.**

***Index Terms*—Bullwhip Effect, Multi-Agent Systems, Supply Chain Resilience, AI Agents, Demand Forecasting, Monte Carlo Tree Search**

I. Introduction

Today's global economy has moved the center of competition away from the market level of individual firms toward the market level of supply chains [23], [91]. Supply chains are not only interconnected, but exposed to instability and disruption that serve as a pressing vulnerability, accentuated by the global breakdown triggered by the COVID-19 pandemic [24], [25], [92]. A main challenge within these networks is the phenomenon called the Bullwhip effect, a chain reaction that occurs when small changes in end-customer demand cause more severe changes in orders and, subsequently, inventory, throughout a supply chain's upstream suppliers [11]–[13]. This chain reaction is wasteful when viewed as inefficient operations, such as the unintended result of excessive purchasing, associated out-of-stocks, and other mismanaged inventory and costs [26], [93]. It follows that supply chain resilience, or the capacity to absorb and adapt to disruption, should be jobs number one for decision makers looking to create and sustain a competitive advantage [27], [72].

While large organizations are taking advantage of advanced data analytics and automation for risk management [5], [94], Micro, Small, and Medium Enterprises (MSMEs) often lack the software, technical expertise, and financial resources to employ them in similar data-based strategies [28], [56]. This disparity creates a significant gap in competitiveness even though MSMEs comprise the backbone of the Indian economy [29], [95]. An industry report issued by NASSCOM identifies this conundrum: "94% of Indian MSMEs believe that AI has the potential to grow their business" yet, "65% do not know the right tools, 59% do not have the budget, and 57% do not have the in-house expertise" to use such AI-based tools [10]. The impetus to conduct this review arises from the need for research synthesizing on AI-based tools that are accessible, low-in-cost, and easy to interpret for MSME entrepreneurs [10], [28]. In order to react to these difficulties, recent research has moved away from conventional centralized planning models, which are typically too rigid for today's dynamic environment, toward artificial intelligence (AI)-enabled decentralized models [30], [31], [57]. Supply chains are now widely accepted as Complex Adaptive Systems (CAS); dynamic networks of many independent agents whose collected behavior is typically indeterminate [14], [32], [41]. As a result, Multi-Agent Systems (MAS) have been developed as an "exceptional and powerful tool" for modeling CAS, as there is a natural analogy between real world supply chain actors and autonomous software agents [8], [42], [43]. Some key technologies within this paradigm include:

- The use of Machine Learning (ML) has advanced predictive analytics and improved the original concepts of

demand forecasting with models such as Random Forest, Gradient Boosting, and Prophet to provide greater accuracy and flexibility [9], [18], [33], [96].

- Large Language Models (LLMs) have developed into autonomous agents that can take disposition actions for inventory with zero-shot while also providing improved explainability with enhanced Chain-of-Thought (CoT) reasoning [1], [17], [97].
- Monte Carlo Tree Search (MCTS) has created inventory optimization functionality to search extremely large state spaces and determine robust ordering policies when conventional approaches have reasonable uncertainty [2], [15], [16], [54].

A recurring theme in the literature is the transition to integrated paradigms that combine some or all of these frameworks into a unified system [4], [34], [98]. The technology-enabled systems utilize the decentralized/autonomous properties of MAS to enable resilient systems, and embed AI models to construct intelligent decisions in the realm of forecasting and optimization [6]. The review aims to distill this literature to appraise the body of existing empirical work on the scope of an AI-Native Multi-Agent System for Micro-, Small- and, Medium Enterprises (MSMEs) by distinguishing the architecture that creates specialized, coordinated agents; Data Harvester; Trend Analyst; Prophet-based Forecaster; and MCTS Optimizer in response to a centralized AI Agent Orchestrator [21], [40]. The plan for this model is low-cost, scalable, and interpretable for purposes of automating supply chain functions from data ingestion all the way through to order fulfillment [10], [28].

The aim of this paper is to examine the literature on AI and Multi-Agent Systems for supply chain management, identify the key technological and practical challenges, and suggest some future areas for consideration from an MSME (micro, small, and medium enterprise) perspective (see table in Appendix A) [36], [37]. The relevant contribution of this paper is providing a roadmap to democratize advanced AI and permit small businesses' better stabilization of processes, reduced Bullwhip effect, and the continued sustainability of competitive growth [35], [55]. The outline of this paper is as follows. Section 2 provides a literature review section that uses structured tables; Section 3 presents the proposed method and architecture of the system; Section 4 seeks to provide critique and discussion of synthesized findings from the literature; Section 5 will outline several research gaps and future research activities; and Section 6 concludes the paper [5], [36].

II. Literature Review

In this segment, we will present the chosen literature in tabular format for intensive examination in order to compare their various main features, including datasets, methods, evaluation metrics, limitations, and relation to the proposed system [36], [37].

Table I of the literature review illustrates a rising trend around intelligent and decentralized supply chains [31], [34], [57]. There are papers that addressed specific aspects of supply chain operations, such as architectural frameworks and optimized algorithms, but altogether they reach a single, powerful statement for an AI system that fits MSMEs ideally. Following this paragraph will be the thematic summary and critique of papers.

III. Methodology of the Review

In this section, we outline the systematic approach for selecting, organizing, and reviewing the literature to help promote transparency and rigor for the review [36]–[38]. This process examined the core issues of our proposed AI-Native Multi-Agent [21], [40].

A. Identifying Literature and Criteria

This literature review identified literature through a systematic search of a number of academic and industrial databases, notably Scopus, Web of Science, IEEE Xplore and arXiv [38], [99]. For a piece of industry-based literature, this review also incorporates a report published by NASSCOM for market context and validation [10].

- **Time frame:** In order to provide a snapshot of contemporary developments related to AI, machine learning, and multi-agent systems in connection to supply chain management; a specific timeframe in the more recent time (2015 through 2025) will be reviewed for coverage [4], [5], [7], [9].
- **Inclusion criteria:** Academic peer-reviewed journals, conference papers, and market-based reports published in the English language were accepted, on the basis that each article covers at least one of the key developments identified in the review—multi-agent systems (MAS), supply chain management (SCM), Bullwhip effect, demand forecasting, inventory optimization, Monte Carlo Tree Search (MCTS), or any applications of AI to the MSME sector [8]–[11], [15], [16], [23], [28], [39], [63].
- **Exclusion criteria:** Other typologies of literature, including book chapters, non-peer-reviewed literature and non-English publications were omitted to maintain consistency of quality specification [37], [64].

B. Categorization and Analytical Approaches

In order to synthesize the outcome from the selected articles, articles were categorized in terms of their meaningful contributions to the research area as follows [61], [65]:

1) **Foundational Frameworks and Reviews:** Articles which provide a high-level overview of the topic, such as what is digital supply chain management (DSCM) and how multi-agent systems (MAS) are used in supply chain management (SCM) [5], [7], [8].
2) **Specific AI models / algorithms:** Articles that document the implementation and context of specific AI models and algorithms, as well as their impact on performance, such as MCTS for inventory management or LLM as an agent [1]–[3], [17].

TABLE I
COMPREHENSIVE LITERATURE REVIEW SUMMARY

Sr.	Paper Title	Dataset Used	Key Features	Models/ Algorithms	Evaluation Parameters	Research Gaps/ Limitations	Relevance to Proposed System
1	InvAgent: A Large Language Model based Multi-Agent System for Inventory Management in Supply Chains [1]	Simulated data across five demand scenarios	Multi-agent LLM system with User Proxy agent; Chain-of-Thought (CoT) prompting	LLMs (GPT-4, GPT-4o, GPT-4-Turbo), Base-Stock, IPPO, MAPPO	Average total reward over 5 simulations	Uses simulated data; high computational cost; plans for RL fine-tuning and real-world testing	Provides architectural precedent for AI Agent Orchestrator; CoT explainability aligns with user transparency objective
2	Artificial intelligence-based inventory management: a Monte Carlo tree search approach [2]	Simulated four-echelon supply chain ("beer game")	First use of MCTS for inventory management; minimizes Bullwhip effect	MCTS vs. GA and RL	Inventory cost and Bullwhip coefficient	Limited to discrete order quantities; assumes perfect information sharing	Justifies MCTS in Agent Optimizer; demonstrates Bullwhip reduction through cost optimization
3	Monte Carlo Graph Search and Transpositions in Monte Carlo Tree Search [3]	Strategic board game (Omega)	Introduces MCGS to handle transpositions via Directed Acyclic Graphs	MCTS and MCGS variants (UCT, UCD)	Win rates, search efficiency	Game-focused; added computational overhead	Guides efficiency improvement of Agent MCTS Optimizer by reducing redundant state evaluations
4	Multi-Agent Deep Reinforcement Learning for Integrated Demand Forecasting and Inventory Optimization [4]	Real-world datasets (Dunnhumby, Favorita, UCI Online Retail)	MARIOD framework integrating demand forecasting and inventory optimization with real-time IoT data	MARL and Transformer networks	MAPE, RMSE, stockouts, service level, turnover, time	Highly complex; limited to simpler supply chain structures	Provides advanced benchmark for comparison; supports modular MSME-friendly approach; suggests roadmap for future deep learning integration
5	Digital Supply Chain Management: A Review and Bibliometric Analysis [5]	1,053 papers from Web of Science	Identifies DSCM technologies (IoT, AI, Blockchain); highlights resilience and risk management	Bibliometric analysis via CiteSpace	Publication trends, keyword and citation mapping	Suggests digital twins as emerging area	Establishes academic context; validates AI-driven SCM; highlights potential future extensions like digital twin integration
6	Enhancing supply chain resilience with multi-agent systems and machine learning [6]	Conceptual case study (electronics firm)	MAS-ML framework with specialized agents; decentralized decision-making	MAS with ML (Random Forest, Gradient Boosting)	Accuracy, lead time, logistics cost	Framework only; lacks implementation	Validates agent-based modular architecture; links it to supply chain resilience and Bullwhip reduction
7	From Agents to Sustainability: Charting the Evolution and Impact of Multi-Agent Systems [7]	281 articles (Scopus, WoS, ScienceDirect)	Confirms MAS as key to managing complex, sustainable supply chains	Bibliometric analysis (VOSviewer)	Trends, clusters, geographical mapping	Scalability and real-time integration challenges	Reinforces MAS as modern paradigm for SCM; positions project within Industry 4.0 sustainability trends
8	Insights on Multi-Agent Systems Applications for Supply Chain Management [8]	Literature survey	Defines supply chains as Complex Adaptive Systems (CAS); formal agent role classification	Review of MAS tools (JADE, Repast, Swarm)	N/A	Lack of unified development methodology	Provides theoretical basis by framing agents as "Functional Agents"; supports decentralized MAS modeling
9	Machine Learning and Deep Learning Models for Demand Forecasting in Supply Chain Management [9]	119 papers from Scopus	Reviews AI/ML/DL forecasting models; notes rise of hybrid approaches	LSTM, RF, SVM, XGBoost	Accuracy, time, complexity, robustness	Data quality and model interpretability issues	Supports use of AI-based forecasting; justifies Prophet model for interpretability and MSME suitability; aligns with Explainable AI goals
10	Unlock AI's Potential for Tech-Enabled MSMEs (NASSCOM & Meta Report) [10]	Sentiment analysis of 220+ MSMEs in India	Highlights AI adoption barriers: awareness, cost, and expertise gaps	N/A	Adoption rates, barrier statistics	Limited to perceptions; lacks technical depth	Provides strong market validation; reinforces need for low-cost, interpretable, and accessible AI solutions for MSMEs—the focus of proposed system

3) **Market context – ways to validate:** Industry reports or inquiries that identify real market challenges and needs, with specific reference to MSMEs [10], [28].

The synthesized literature review findings are shared in Section 4 in an appropriate and condensed table that compares all articles in terms of participant methodology, innovativeness evaluative criteria, and research literature gaps [36], [66].

C. Proposed System Architecture

The goal was to develop a proposed architecture for an AI-Native Multi-Agent System addressing MSME sector challenges [6], [28]. The proposed architecture features an AI Agent Orchestrator as a central hub overseeing dedicated autonomous agents engaged in their own state-driven control loops, as illustrated in Figure 1 [21], [40].

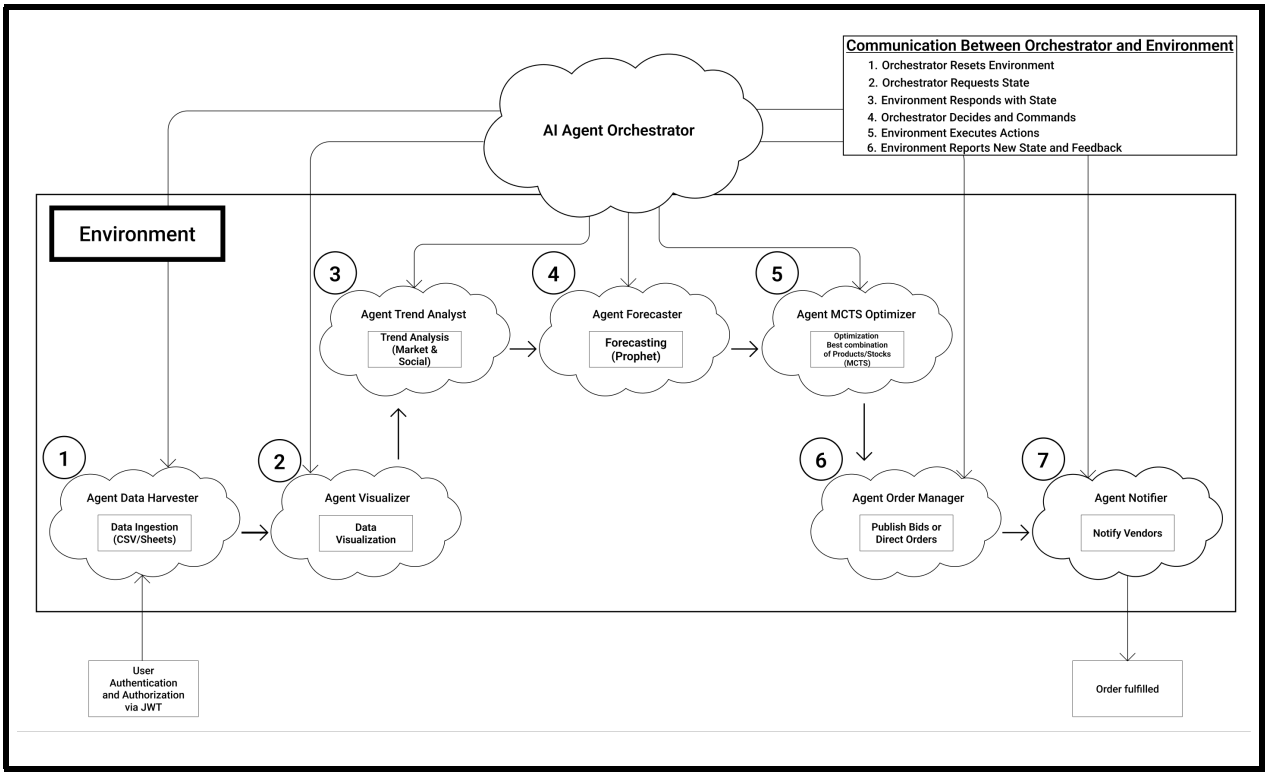

Fig. 1. AI-Native Multi-Agent Architecture for Supply Chain Management

The proposed architecture invites the following agents:

- **Agent Data Harvester** — will be able to ingests data from popular MSME formats such as CSV or Google sheets [58].
- **Agent Trend Analyst** — will extracts actionable insights from market and social media data such as Twitter [59], [68].
- **Agent Forecaster** — will perform a model demand prediction using the Prophet model [18].
- **Agent MCTS Optimizer** — will use Monte Carlo Tree Search methods to better inform good inventory/re-orders decisions [2], [15], [16].
- **Operational Agents** — will automate engagements and coordination of communications to suppliers even to the operational level including an Agent Order Manager and Agent Notifier [57], [69].

Overall, the intent of the proposed model was practical/integrated, low cost, and interpretable builds to automate operational decisions associated moderate bullwhip effect [11], [60]. Overall, the conservation presents an authentic learning context for the application of technologies and assessments of important architectures with regards to the at risk literature review presented [5], [9].

IV. Thematic Synthesis of the Literature

This part structures and synthesizes the important results of the literature into four central themes of the review: the emergence of Multi-Agent Systems (MAS) as a new paradigm for supply chain management; the development of AI-enabled forecasting; the use of advanced optimization techniques, such as Monte Carlo Tree Search (MCTS); and the critical, market-validated desire for easy access to AI tools for MSMEs [2], [8]–[10].

A. Theme 1: The Rise of Multi-Agent Systems as a Dominant Paradigm for SCM

A common theme that arises from the literature is the understanding that the modern supply chains have shifted from simple, linear chains to Complex Adaptive Systems (CAS) [14], [32], [41]. A CAS is a collection of many autonomous agents in a dynamic network that interact, which generates emergent behavior that cannot be predicted [41], [70]. The complexity of CAS renders traditional, centralized management models ineffective [30]. As a result, Multi-Agent Systems (MAS) have been classified as an "excellent and powerful tool" for supply chain modeling [8], [31].

What Is Known: Real-world actors in a supply chain (e.g., suppliers, distributors) correlate naturally to software agents in a MAS model; thus, it may be possible to take a bottom-up modeling approach that captures the decentralized properties of a real-world supply chain system [42], [43], [73]. Since 2021, research on using MAS to model both sustainable and circular supply chains has increased rapidly in alignment with the conceptual practices of Industry 4.0 [7], [20], [71].

Key Findings: Owing to the ability to allow for decentralized and autonomous decision-making, MAS architecture provides the system with the ability to be flexible and adaptive, resulting

in increased resilience in the supply chain [24], [31], [57], [72]. The agents in the system can be formally classified by their roles as Enterprise, Resource, or Functional; the specialist agents in the proposed system (Agent Forecaster, Agent MCTS Optimizer) are Functional Agents [8], [40].

Limitations and Gaps: While MAS possess advantageous characteristics for widespread adoption, several issues surround applying MAS to practical models, including scalability, real-time data integration, and socio-economic factors [7], [44]. Moreover, the literature notes there is no universal methodology for developing MAS models, indicating a need for more streamlined methods or frameworks suitable for broad adoption [8], [40].

B. Theme 2: AI-Driven Forecasting: From Statistical Models to Advanced Deep Learning

The existing body of research validates an overarching and significant shift in the context of traditional statistical forecasting methods (i.e., ARIMA) to more agile and accurate forecasting models associated with artificial intelligence [9], [45], [100]. This shift in the forecasting landscape illustrates that traditional models are often obsolete in terms of recognizing how significantly consumer behaviors have shifted, and the marked volatility seen in the supply chain [46].

What is Known: AI provides a 'disruptive alternative' compared to traditional forecasting models with improved accuracy and agility. Research activity in this area rapidly accelerated post 2021 and there has been clear evidence of machine learning (ML) and deep learning (DL) models being introduced [9], [75], [76].

Key Findings: More advanced models have proven more accurate. For example, Long Short-Term Memory (LSTM) networks capture 15-20% fewer forecast errors than ARIMA (classical statistical) forecasts [47], [48]. Another strong trend is hybrid models (e.g., SARIMA-LSTM-BP) and ensemble models that borrow the strengths of statistical and deep learning models to achieve even greater accuracy and reduction in forecasting errors. More recently, very sophisticated architectures (e.g., Transformers) have been developed that can process multiple data streams at once (e.g., historical sales, promotions, real-time sensor feedback) resulting in improved performance [4], [49], [50].

Limitations and Gaps: A primary and recurring limitation in many state-of-the-art DL models is the notion of interpretability: the "black box" dilemma. This is detrimental for users not acquainted with MLE, preventing them from trusting or acting on the model predictions. This gap has contributed to further growing conversations of the need for Explainable AI (XAI) [51], [52], [77], [78]. Further, the performance of some of these models is also contingent upon the quality of data and perception of accessibility of data, which remains a significant and practical limitation as well [53], [79].

C. Theme 3: Advanced Optimization Techniques: The Role of Monte Carlo Tree Search (MCTS)

For complicated inventory optimization models, it is usually not possible to find the best ordering policy via analytical techniques, because of the size and uncertainty associated with complex models. Research has demonstrated that heuristic AI-based techniques have more potential to solve this type of problem [19], [39], [63].

What is Known: MCTS, which is commonly used in AI in games, is a powerful technique used for sequential decision problems which can manage extremely large state spaces regardless of the need to evaluate every possibility [15], [16], [101]. While its application to supply chain inventory management is novel, it fits the problem exceptionally well [2].

Key Findings: The performance of MCTS has produced exceptional results. One of the key findings is a dynamic order policy determined through MCTS can successfully eliminate the Bullwhip effect, as a result of minimizing systemwide inventory cost. In simulated beer game scenarios, MCTS significantly outperformed previous AI methods such as Genetic Algorithms (GA) and Reinforcement Learning (RL) [2], [22]. Furthermore, we can make the MCTS core algorithm more efficient by implementing a graph-based structure (Monte Carlo Graph Search) to manage transpositions and reduce state evaluation overhead, yielding better performance [3].

Limitations and Gaps: The conventional MCTS technique is not necessarily suitable for continuous (i.e., non-integer) order quantities, since the action space is usually limited to discrete sets of values. MCTS's use in SCM is still nascent. Most of the work has been performed in simulation studies, lacking validation on real-world operations and the use of the products brought to market [2], [54].

D. Theme 4: The Critical Need for Accessible AI Solutions for MSMEs

Perhaps the most interesting observation is that there is a proven, market-validated, and real-world need for solutions of the type discussed above. While scholarship can occasionally discuss the most recent evolution in technologies being used, the reports from the traditional industry also demonstrate the stark difference between the promise of AI technologies and the actual use of these technologies by MSMEs [10].

What is Known: MSMEs are an important part of sectors of the economy in both global and Indian contexts but are at a competitive disadvantage based on their lack of access to the advanced automation and data analytic tools that benefit larger organizations [29], [56], [95].

Key Findings: A recent groundbreaking (NASSCOM & Meta) report provides compelling empirical evidence to support this in an Indian context. The report found that while 94% of Indian MSMEs believe in the potential of A.I. to grow their business, they encountered 3 key barriers: 1) Lack of awareness of the correct tools (65%); 2) Budget restrictions or limits (59%); and 3) Lack of internal technical expertise (57%). Therefore, there are clearly viable and substantial market opportunities for appropriate low-cost, user-friendly, and easily interpretable solutions for non-technical users [10], [55], [102].

Limitations and Gaps: The "gap" established here is not present in the academic literature, it is in the marketplace. The report also found that 45% of MSME surveyed experienced the requirement for industry-specific use cases to aid their understanding of AI. This is confirming factors field that there is a severe gap in AI tools and platforms that are developed from the beginning to address the economic and practical reality of small to medium sized businesses [10], [28].

V. Critical Analysis & Synthesis: A Discussion

The literature reviewed articulates an unambiguous and commonly understood path in supply chain management: there is a movement towards decentralized, intelligent, and resilient systems [23], [24]. Although each study focuses on a different component—be it systematic architectures or optimal forecasting models—synthesizing the studies presents patterns, methodological dialogues, and implications for future work, which collectively make a compelling case for an integrated solution for MSMEs [2], [8]–[10].

A. Comparing Architectural Approaches: The Consensus on Multi-Agent Systems

A common thread woven within the literature is the consensus that MAS is the most suitable architectural framework for supply chains today [8], [31], [57]. Older, centralized models are not designed for the unpredictable and adaptive nature of contemporary market conditions [30]. In examining supply chains as Complex Adaptive Systems (CA), the authors argue that there is only one appropriate framework to properly describe their emergent behavior, which simply put would be decentralized, "bottom-up" agents [14], [32], [41].

Methodological Strengths: The primary strength of MAS is that they correspond more closely with real supply chains in that there are independent agents (suppliers, distributors, etc.) each making independent, seemingly-coordinated decisions, which creates a superior level of adaptability and resiliency [24], [42], [72], [73]. Papers proposing MAS frameworks, such as noted with Farazi's proposal and others, can further validate the modular nature of our proposed model with specialized agents, referred to here as "Functional Agents," assigned commons will be given to the specific areas of required work, e.g. forecasting, managing inventory, etc [6], [40].

Weaknesses and Underexplored Areas: The most apparent weakness identified many of the MAS frameworks proposed is that they are conceptual frameworks or validated via SIMULATIONS [44]. To illustrate my point, InvAgent shows, in theory, that LLM agents work, however the data it produces is entirely simulated, and it is cumbersome in both computational and processing power, as a follow up point though means it does not support low resource MSMEs practical counter-argument, this highlights the division between frameworks of MAS that are supported theoretically, and reasonable and plausible operational methods for those businesses [1], [10], [28].

B. The Forecasting Dilemma: Contrasting Accuracy with Interpretability

Another obvious trend is the rapid advancement of demand forecasting models, moving away from traditional statistical models to complex artificial intelligence models [9], [45]. The available literature is progressively moving toward more and more complicated and accurate models moving from machine leaning ensemble statistical models, like Random Forests, to advanced ensemble-deep learning architectures like LSTMs and Transformers [48], [50], [96].

Contradictions and Debates: The inconsistency illustrates an important debate in the literature - the tradeoff between accuracy and interpretability [51], [77]. Although advanced models such as LSTMs and Transformers may achieve advanced state-of-the-art accuracy because of their ability to input large and multimodal datasets, their layered architectures often produce "black box" problems - meaning that the rationale for any prediction remains largely "unknowable" [4], [52], [78]. Black-box-issues in models obstruct them from being used, in practice, especially for non-technical-users - who typically require some degree of trust before they can justify using a systems behavioral guidance for their high stakes business decisions [90].

Justifying the Proposed Approach: This incongruity strongly advocates for the Prophet model as the best choice to incorporate into our system [18]. While it does not have the magnetic allure of being powered by a large transformer model, the Prophet model is specialized in providing business-focussed forecasting of time-series data and tends to perform well in low-data / high-variation business environments, as is the case with micro, small and medium-sized enterprises (MSMEs). The Prophet model allows for interpretability, which that explicit method choice embodies trust and engagement of the user even though it might not be achieving the highest predictive accuracy or sacrificing interpretability and emphasizing a critical goal of the project at the outset [46].

C. A Breakthrough in Optimization: The Emergence of MCTS

The literature recognizes Monte Carlo Tree Search (MCTS) as an innovative approach to addressing the challenging problem of inventory optimization. The utilization of MCTS is superior to other AI methods like Genetic Algorithms (GA) and Reinforcement Learning (RL) [2], [22].

Methodological Strengths: MCTS's primary advantage, included with its ability to penetrate extremely large and complex state spaces more effectively than other AI methods without the curse of dimensionality often associated with state-based models, is increased fidelity in modeling the environment of the supply chain [15], [16]. The literature provides compelling evidence that MCTS also effectively eliminates the Bullwhip effect as a secondary effect of reducing inventory costs, which is relevant to the central problem considered in the ideation of our project [2]. Lastly, the literature provides a plausible pathway for methodological improvement in utilizing Monte Carlo Graph Search (MCGS) as a deft approach to

increase algorithmic efficiency and facilitate chaining repeated calculations throughout the execution of MCTS [3].

Underexplored Areas: The application of MCTS into supply chain management is a young but developed research area that has yet to be fully explored. The majority of the literature study the experiments, and in fact the entirety, including the foundational literature is more in the form of simulation 'beer game' patterns [2], [13]. Hence, there is a unique and warranted value in carrying out and popping the MCTS design within a practical, data-driven construct using real business world data [5].

D. Synthesis: Bridging the Gap Between Advanced Research and Market Reality

The most notable takeaway emerges by comparing cutting-edge academic systems with MSME market realities: a stark access gap persists [10], [28]. At one end, integrated, computationally expensive frameworks like MARIOD jointly optimize forecasting and inventory using MARL and Transformers [4]. At the other end, NASSCOM documents that most MSMEs cannot access such technologies due to awareness (65%), cost (59%), and talent (57%) constraints [10]. This contradiction highlights the need for solutions that are powerful yet accessible.

The proposed AI-Native Multi-Agent System addresses this gap by combining proven, usable technologies into a low-cost platform:

- A flexible, resilient MAS architecture [8], [31].
- An interpretable forecasting model (Prophet) that balances accuracy and trust [18].
- A powerful optimization algorithm (MCTS) that directly targets the Bullwhip effect [2].

Packaging these components into a single, user-friendly system directly responds to industry's call for cost-effective, explainable AI that helps MSMEs stabilize operations and grow competitively [35], [55].

VI. Future Directions and Identified Research Gaps

In sum, our synthesis of the current literature indicates that significant progress has been made regarding the application of AI and Multi-Agent Systems towards supply chain management; however, substantially important research gaps and emerging opportunities still exist [5], [8], [9]. In this concluding section, we propose plausible avenues for research that leverage the existing body of knowledge toward improved intelligent, resilient, and accessible supply chain solutions [23], [24].

A. Enhancing Agent Intelligence and Autonomy

One key direction for future research is to improve how intelligent and reasoning agents behave in MAS [40], [67].

a) Combination of Deep Learning and Hybrid Forecasting Models: The literature suggests forecasting models will continue to become more advanced [9], [46]. While the proposed system uses Prophet in part due to its interpretability, future work should develop hybrid models that integrate the strengths of statistical methods with deep learning [49]. Subsequent versions of the Agent Forecaster can incorporate architectures like Long Short-Term Memory (LSTM) networks and Transformers to capture complex, non-linear demand patterns, as advocated by several studies [4], [47], [48], [50].

b) Exploration of Reinforcement Learning (RL) for Optimization: While the complete review shows MCTS is an optimization tool, several papers identify the next frontier to be in Reinforcement Learning (RL) and Multi-Agent Reinforcement Learning (MARL) [4], [19], [80]. An emerging opportunity could be to adapt the Agent MCTS Optimizer to develop into a learning-based agent, that through trial and error can develop a dynamic ordering policy. Furthermore, tuning LLM-based agents through reinforcement [80], [81].

B. Expanding System Architecture and Data Integration

Beyond individual agent capabilities, significant opportunities exist in evolving the overall system architecture to be more integrated and data-aware [5], [20].

a) Development of Digital Twin Integration: One of the key emerging opportunities in the literature is the concept of the digital twin [82], [83]. Recommendations for future research would advance this idea by developing an Agent Digital Twin that could develop a virtual real-time representation of an MSME's supply chain, effectively creating a baseline for the Agent MCTS Optimizer to conduct more realistic and complex simulation scenarios supporting predictive risk management and can be stress tested against a variety of potential disruptions [24].

b) Incorporating Real-Time IoT Sensor Data: The MARIOD framework points to the possible advantage of combining historic data with IoT-based sensor measurements, RFID, and smart shelves [4], [84], [85]. For systems targeting the MSME size of firms, one of the main research gaps is to develop low-cost approaches to bring such data streams into the system. This would allow the MCTS Optimizer to change from predictive to prescriptive analytics which could respond in realtime to a real-world event, such as a sudden drop in on-shelf inventory [85].

c) Transitioning to Jointly Optimized Frameworks: Current systems, including the architecture we propose here, generally operate through some variant of a linear sequential path (i.e., forecasting first, and then optimizing) [39]. In MARIOD, we proposed a sophisticated jointly optimized framework where forecasting and operational management learn together [4]. A challenge characterized by the field is how to co-design such frameworks without sacrificing modularity, computational efficiency, or accessibility to MSMEs [35].

C. Broadening the Application Scope and Impact

Finally, future work should aim to broaden the application of these systems to address wider societal and economic needs [20], [103].

a) Addressing Sustainability and the Circular Economy: Several review papers have identified sustainability, reverse logistics, and the circular economy as important and underdeveloped areas for MAS [7], [86], [87]. The modularity of a MAS architecture fits well to facilitate this expansion. For example, new agents could be developed to manage product returns, improve recycling, or assess the carbon footprint of logistics, thus extending the impact of the system beyond operational efficiency to an environmental responsibility element [88].

b) Expanding to Other Sectors: While the future of the research is frequently positioned in manufacturing and retail, the MAS-ML framework could be modified to engage with other sectors critical to the MSME economy, including healthcare, transportation, and agriculture [6], [89]. Each of these sectors presents its own unique challenges that would necessitate specialized agents and models, representing an exciting avenue for future applied researchers.

c) Deepening Research into Explainable AI (XAI): The "black box" problem is still one of the major hurdles in the adoption of AI within MAS [51], [52], [77]. Increasingly complex forecasting and optimization models demand effective XAI techniques to fully understand and trust the results of the models they will rely on. Future research should not only focus on accuracy improvements in the models, but also on the quest for new ways to provide decision outcomes that are transparent, auditable, and can be trusted by the non-technical users of these models [78], [90].

VII. Conclusion

This article has reviewed recent literature linking artificial intelligence, multi-agent systems, and supply chain management, which has yielded several key findings [5], [8]. It is realistic to say there is significant publishing activity in this area—and rightfully so, since today's supply chains are Complex Adaptive Systems (CAS), and Multi-Agent Systems (MAS) was developed as a means to manage Complex Adaptive Systems in the context of supply chain management—using a decentralized and resilient structure to manage supply chain operations [14], [31], [41]. Similarly, while the advent of artificial intelligence has manifested into improved demand forecasting outcomes, as a result of greater predictive accuracy associated with deep learning models, the balance between predictive accuracy and model performance interpretability is a continuing concern [9], [52], [77]. Finally, through the critical examination of the literature, we noted the emergence of Monte Carlo Tree Search (MCTS) as a new and beneficial optimization tool, which may help address as well the Bullwhip effect, as it relates to optimization, without regard for the fact that the optimization itself is through cost [2], [60].

The importance of this theme cannot be overstated. Creating resilient supply chains is critical to the stability of the economy. Furthermore, the Bullwhip effect is something we can fix to impact operational efficiencies and profits directly [11], [23]. This is particularly problematic for Micro, Small and Medium Enterprises (MSMEs) which serve as the backbone of the Indian economy and are generally the most blamed for the increased volatility in the supply chain [29], [95]. A report by NASSCOM indicated a substantial market opportunity highlighting that many MSMEs had financial constraints to access artificial intelligence, low levels of technical knowledge and a general lack of knowledge to access the right tools [10].

This review makes an important contribution to the field in three ways. First, it collates and synthesizes knowledge from different research areas, in which ways have included architectural theory, and different specifics in AI algorithms as well as real-world market research, into a coherent framework, providing a compelling case for a theoretical MSME-specific integrated solution [5], [8]. Second, it presents clarity to a primary debate in the field by exploring the tension between being academic and having accuracy versus being a business and needing something that is intelligible, and thus legitimizing the thoughtful selection of an accessible and intelligible AI [18], [46]. Finally, the review highlights noteworthy gaps to research specifically, and opportunities to study, the integration of digital twins and sustainability, and more broadly acknowledges the need for Explainable AI, therefore helping to outline a twenty-first-century research agenda [7], [52], [78], [82], [86], [90]. The review creates a bridge to the gap in scholarly theoretical work and thoughtfully meeting the needs of the largest constituencies of the business community, as a case for scholars to pivot their research agenda towards creating AI systems that are not only powerful, but accessible, practical and impactful.

References

[1] Y. Quan and Z. Liu, "InvAgent: A Large Language Model based Multi-Agent System for Inventory Management in Supply Chains," *arXiv preprint arXiv:2407.11384*, Jul. 2024.

[2] D. Preil and M. Krapp, "Artificial intelligence-based inventory management: a Monte Carlo tree search approach," *Annals of Operations Research*, vol. 308, pp. 415–439, Jan. 2022.

[3] A. Mulvey Izquierdo, "Monte Carlo Graph Search and Transpositions in Monte Carlo Tree Search," *Preprint*, Nov. 2024.

[4] Y. Yang, M. Wang, J. Wang, P. Li, and M. Zhou, "Multi-Agent Deep Reinforcement Learning for Integrated Demand Forecasting and Inventory Optimization in Sensor-Enabled Retail Supply Chains," *Sensors*, vol. 25, no. 8, p. 2428, Apr. 2025.

[5] H. Zhang, Y. Lv, S. Zhang, and Y. D. Liu, "Digital Supply Chain Management: A Review and Bibliometric Analysis," *Journal of Global Information Management*, vol. 32, no. 1, 2021.

[6] M. Z. R. Farazi, "Enhancing supply chain resilience with multi-agent systems and machine learning: a framework for adaptive decision-making," *The American Journal of Engineering and Technology*, vol. 7, no. 3, pp. 6–20, Mar. 2025.

[7] B. Elghomri, F. Messaoudi, H. Allaoui, and A. El Amraoui, "From Agents to Sustainability: Charting the Evolution and Impact of Multi-Agent Systems in Circular Supply Chain Management," in *2025 International Conference on Circuit, Systems and Communication (ICCSC)*, 2025.

[8] R. Dominguez and S. Cannella, "Insights on Multi-Agent Systems Applications for Supply Chain Management," *Sustainability*, vol. 12, no. 5, p. 1935, Mar. 2020.

[9] K. Douaioui, R. Oucheikh, O. Benmoussa, and C. Mabrouki, "Machine Learning and Deep Learning Models for Demand Forecasting in Supply Chain Management: A Critical Review," *Applied System Innovation*, vol. 7, no. 5, p. 93, Sep. 2024.

[10] NASSCOM and Meta, "Unlock AI's Potential for Tech-Enabled MSMEs," Industry Report, 2024.

[11] H. L. Lee, V. Padmanabhan, and S. Whang, "Information distortion in a supply chain: The bullwhip effect," *Management Science*, vol. 43, no. 4, pp. 546–558, 1997.

[12] J. W. Forrester, *Industrial Dynamics*. Cambridge, MA: MIT Press, 1961.

[13] J. D. Sterman, "Modeling managerial behavior: Misperceptions of feedback in a dynamic decision making experiment," *Management Science*, vol. 35, no. 3, pp. 321–339, 1989.

[14] T. Y. Choi, K. J. Dooley, and M. Rungtusanatham, "Supply networks and complex adaptive systems: Control versus emergence," *Journal of Operations Management*, vol. 19, no. 3, pp. 351–366, 2001.

[15] L. Kocsis and C. Szepesvári, "Bandit based Monte-Carlo planning," in *European Conference on Machine Learning*, 2006, pp. 282–293.

[16] C. B. Browne *et al.*, "A survey of Monte Carlo tree search methods," *IEEE Transactions on Computational Intelligence and AI in Games*, vol. 4, no. 1, pp. 1–43, 2012.

[17] B. Li, K. Mellou, B. Zhang, J. Pathuri, and I. Menache, "Large language models for supply chain optimization," *arXiv preprint arXiv:2307.03875*, 2023.

[18] S. J. Taylor and B. Letham, "Prophet: forecasting at scale," *The American Statistician*, vol. 72, no. 1, pp. 37–45, 2018.

[19] B. Rolf, L. Jackson, M. Müller, S. Lang, T. Reggelin, and D. Ivanov, "A review on reinforcement learning algorithms and applications in supply chain management," *International Journal of Production Research*, vol. 61, no. 21, pp. 7151–7179, 2023.

[20] H. Lasi, P. Fettke, H.-G. Kemper, T. Feld, and M. Hoffmann, "Industry 4.0," *Business & Information Systems Engineering*, vol. 6, no. 4, pp. 239–242, 2014.

[21] Q. Wu *et al.*, "Autogen: Enabling next-gen llm applications via multi-agent conversation framework," *arXiv preprint arXiv:2308.08155*, 2023.

[22] S. K. Chaharsooghi, J. Heydari, and S. H. Zegordi, "A reinforcement learning model for supply chain ordering management: An application to the beer game," *Decision Support Systems*, vol. 45, no. 4, pp. 949–959, 2008.

[23] M. Christopher and M. Holweg, "Supply Chain 2.0 revisited: a framework for managing volatility-induced risk in the supply chain," *International Journal of Physical Distribution & Logistics Management*, vol. 47, no. 1, pp. 2–17, 2016.

[24] D. Ivanov and A. Dolgui, "Viability of intertwined supply networks: extending the supply chain resilience angles towards survivability," *International Journal of Production Research*, vol. 58, no. 10, pp. 2904–2915, 2020.

[25] S. Singh, R. Kumar, R. Panchal, and M. K. Tiwari, "Impact of COVID-19 on logistics systems and disruptions in food supply chain," *International Journal of Production Research*, vol. 59, no. 7, pp. 1993–2008, 2021.

[26] F. Chen, Z. Drezner, J. K. Ryan, and D. Simchi-Levi, "Quantifying the bullwhip effect in a simple supply chain," *Management Science*, vol. 46, no. 3, pp. 436–443, 2000.

[27] S. Y. Ponomarov and M. C. Holcomb, "Understanding the concept of supply chain resilience," *The International Journal of Logistics Management*, vol. 20, no. 1, pp. 124–143, 2009.

[28] S. Mittal, M. A. Khan, D. Romero, and T. Wuest, "Smart manufacturing: Characteristics, technologies and enabling factors," *Proceedings of the Institution of Mechanical Engineers, Part B: Journal of Engineering Manufacture*, vol. 233, no. 5, pp. 1342–1361, 2018.

[29] S. Kumar, R. K. Pandey, and A. Sharma, "Artificial intelligence in Indian MSMEs: Opportunities and challenges," *International Journal of Business Innovation and Research*, vol. 14, no. 2, pp. 234–250, 2017.

[30] C. Schneeweiss, "Distributed decision making in supply chain management," *International Journal of Production Economics*, vol. 84, no. 1, pp. 71–83, 2003.

[31] W. Shen, Q. Hao, H. J. Yoon, and D. H. Norrie, "Applications of agent-based systems in intelligent manufacturing: An updated review," *Advanced Engineering Informatics*, vol. 20, no. 4, pp. 415–431, 2006.

[32] S. D. Pathak, J. M. Day, A. Nair, W. J. Sawaya, and M. M. Kristal, "Complexity and Adaptivity in Supply Networks," *Decision Sciences*, vol. 38, no. 4, pp. 547–580, 2007.

[33] R. Carbonneau, K. Laframboise, and R. Vahidov, "Application of machine learning techniques for supply chain demand forecasting," *European Journal of Operational Research*, vol. 184, no. 3, pp. 1140–1154, 2008.

[34] H. Min, "Artificial intelligence in supply chain management: theory and applications," *International Journal of Logistics Research and Applications*, vol. 13, no. 1, pp. 13–39, 2010.

[35] J. Mueller, "Enabling innovation in SMEs through big data analytics and artificial intelligence," *Journal of Business Research*, vol. 100, pp. 155–162, 2019.

[36] D. Tranfield, D. Denyer, and P. Smart, "Towards a methodology for developing evidence-informed management knowledge," *British Journal of Management*, vol. 14, no. 3, pp. 207–222, 2003.

[37] B. Kitchenham and S. Charters, "Guidelines for performing systematic literature reviews in software engineering," Tech. Rep., Keele University and Durham University, 2007.

[38] A. Fink, *Conducting Research Literature Reviews*, 5th ed. Thousand Oaks, CA: SAGE Publications, 2019.

[39] E. A. Silver, D. F. Pyke, and D. J. Thomas, *Inventory and Production Management in Supply Chains*, 4th ed. Boca Raton, FL: CRC Press, 2016.

[40] M. Wooldridge, *An Introduction to MultiAgent Systems*, 2nd ed. Chichester, UK: John Wiley & Sons, 2009.

[41] J. H. Holland, "Complex adaptive systems," *Daedalus*, vol. 121, no. 1, pp. 17–30, 1992.

[42] M. S. Fox, M. Barbuceanu, and R. Teigen, "Agent-oriented supply-chain management," *International Journal of Flexible Manufacturing Systems*, vol. 12, pp. 165–188, 2000.

[43] C. M. Macal and M. J. North, "Tutorial on agent-based modeling and simulation," in *Proceedings of the Winter Simulation Conference*, 2005, pp. 2–15.

[44] H. V. D. Parunak, "What can agents do in industry, and why?" in *Cooperative Information Agents II*, Berlin: Springer, 1998, pp. 1–18.

[45] J. S. Armstrong, ed., *Principles of Forecasting*. Boston, MA: Kluwer Academic Publishers, 2001.

[46] S. Makridakis, E. Spiliotis, and V. Assimakopoulos, "Statistical and Machine Learning forecasting methods," *PLoS ONE*, vol. 13, no. 3, p. e0194889, 2018.

[47] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

[48] K. Bandara, C. Bergmeir, and S. Smyl, "Forecasting across time series databases using recurrent neural networks," *Expert Systems with Applications*, vol. 140, p. 112896, 2020.

[49] S. Smyl, "A hybrid method of exponential smoothing and recurrent neural networks," *International Journal of Forecasting*, vol. 36, no. 1, pp. 75–85, 2020.

[50] A. Vaswani *et al.*, "Attention is all you need," in *Advances in Neural Information Processing Systems*, 2017, pp. 5998–6008.

[51] A. B. Arrieta *et al.*, "Explainable Artificial Intelligence (XAI): Concepts and challenges," *Information Fusion*, vol. 58, pp. 82–115, 2020.

[52] A. Adadi and M. Berrada, "Peeking inside the black-box: A survey on Explainable Artificial Intelligence," *IEEE Access*, vol. 6, pp. 52138–52160, 2018.

[53] B. T. Hazen, C. A. Boone, J. D. Ezell, and L. A. Jones-Farmer, "Data quality for data science and big data," *International Journal of Production Economics*, vol. 154, pp. 72–80, 2014.

[54] R. Coulom, "Efficient selectivity and backup operators in Monte-Carlo tree search," in *Computers and Games*, Berlin: Springer, 2007, pp. 72–83.

[55] B. Ramdani, P. Kawalek, and O. Lorenzo, "Predicting SMEs' adoption of enterprise systems," *Journal of Enterprise Information Management*, vol. 22, pp. 10–24, 2009.

[56] J. Thakkar, A. Kanda, and S. G. Deshmukh, "Supply chain management in SMEs," *Asia Pacific Journal of Marketing and Logistics*, vol. 20, no. 1, pp. 97–131, 2008.

[57] N. R. Jennings, "On agent-based software engineering," *Artificial Intelligence*, vol. 117, no. 2, pp. 277–296, 2000.

[58] H. Chen, R. H. L. Chiang, and V. C. Storey, "Business intelligence and analytics: From big data to big impact," *MIS Quarterly*, vol. 36, no. 4, pp. 1165–1188, 2012.

[59] B. K. Chae, "Insights from hashtag #supplychain and Twitter analytics," *International Journal of Production Economics*, vol. 165, pp. 247–259, 2015.

[60] J. Dejonckheere, S. M. Disney, M. R. Lambrecht, and D. R. Towill, "Measuring and avoiding the bullwhip effect," *European Journal of Operational Research*, vol. 147, pp. 567–590, 2003.

[61] V. Braun and V. Clarke, "Using thematic analysis in psychology," *Qualitative Research in Psychology*, vol. 3, no. 2, pp. 77–101, 2006.

[62] S. Chopra and P. Meindl, *Supply Chain Management*, 3rd ed. Upper Saddle River, NJ: Pearson, 2007.

[63] P. H. Zipkin, *Foundations of Inventory Management*. Boston, MA: McGraw-Hill, 2000.

[64] K. Petersen, R. Feldt, S. Mujtaba, and M. Mattsson, "Systematic mapping studies in software engineering," in *EASE Conference*, 2008, pp. 68–77.

[65] J. Thomas and A. Harden, "Methods for the thematic synthesis of qualitative research," *BMC Medical Research Methodology*, vol. 8, p. 45, 2008.

[66] D. M. Rousseau, J. Manning, and D. Denyer, "Evidence in management and organizational science," *Academy of Management Annals*, vol. 2, no. 1, pp. 475–515, 2008.

[67] S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 3rd ed. Upper Saddle River, NJ: Prentice Hall, 2010.

[68] B. Pang and L. Lee, "Opinion mining and sentiment analysis," *Foundations and Trends in Information Retrieval*, vol. 2, pp. 1–135, 2008.

[69] G. Weiss, ed., *Multiagent Systems*. Cambridge, MA: MIT Press, 1999.

[70] P. W. Anderson, K. J. Arrow, and D. Pines, eds., *The Economy as an Evolving Complex System*. Reading, MA: Addison-Wesley, 1988.

[71] H. Kagermann, W. Wahlster, and J. Helbig, "Recommendations for implementing INDUSTRIE 4.0," Forschungsunion, 2013.

[72] Y. Sheffi, *The Resilient Enterprise*. Cambridge, MA: MIT Press, 2005.

[73] J. M. Swaminathan, S. F. Smith, and N. M. Sadeh, "Modeling supply chain dynamics: A multiagent approach," *Decision Sciences*, vol. 29, no. 3, pp. 607–632, 1998.

[74] E. Bonabeau, "Agent-based modeling: Methods and techniques," *Proceedings of the National Academy of Sciences*, vol. 99, pp. 7280–7287, 2002.

[75] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, pp. 436–444, 2015.

[76] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA: MIT Press, 2016.

[77] Z. C. Lipton, "The mythos of model interpretability," *Queue*, vol. 16, no. 3, pp. 31–57, 2018.

[78] C. Molnar, *Interpretable Machine Learning*, 2020.

[79] R. Y. Wang and D. M. Strong, "Beyond accuracy: What data quality means to data consumers," *Journal of Management Information Systems*, vol. 12, no. 4, pp. 5–33, 1996.

[80] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. Cambridge, MA: MIT Press, 2018.

[81] L. Ouyang *et al.*, "Training language models to follow instructions with human feedback," in *NeurIPS*, 2022, pp. 27730–27744.

[82] M. Grieves and J. Vickers, "Digital twin: Mitigating unpredictable behavior in complex systems," in *Transdisciplinary Perspectives on Complex Systems*, Cham: Springer, 2017, pp. 85–113.

[83] F. Tao, H. Zhang, A. Liu, and A. Y. C. Nee, "Digital twin in industry: State-of-the-art," *IEEE Transactions on Industrial Informatics*, vol. 15, no. 4, pp. 2405–2415, 2019.

[84] L. Atzori, A. Iera, and G. Morabito, "The Internet of Things: A survey," *Computer Networks*, vol. 54, no. 15, pp. 2787–2805, 2010.

[85] J. Lee, B. Bagheri, and H.-A. Kao, "A cyber-physical systems architecture for Industry 4.0," *Manufacturing Letters*, vol. 3, pp. 18–23, 2015.

[86] K. Govindan and H. Soleimani, "A review of reverse logistics and closed-loop supply chains," *Journal of Cleaner Production*, vol. 142, pp. 371–384, 2017.

[87] J. Sarkis, Q. Zhu, and K. Lai, "An organizational theoretic review of green supply chain management," *International Journal of Production Economics*, vol. 130, no. 1, pp. 1–15, 2011.

[88] S. Seuring and M. Müller, "From a literature review to a conceptual framework for sustainable supply chain management," *Journal of Cleaner Production*, vol. 16, no. 15, pp. 1699–1710, 2008.

[89] R. Dubey, A. Gunasekaran, S. J. Childe, T. Papadopoulos, and Z. Luo, "Big data analytics and organizational culture," *International Journal of Production Economics*, vol. 210, pp. 120–136, 2019.

[90] M. T. Ribeiro, S. Singh, and C. Guestrin, "'Why should I trust you?': Explaining predictions," in *ACM SIGKDD*, 2016, pp. 1135–1144.

[91] M. E. Porter, *Competitive Advantage: Creating and Sustaining Superior Performance*. New York, NY: Free Press, 1985.

[92] R. B. Handfield, J. V. Finkenstadt, S. A. Schneller, A. Godfrey, and P. Guinto, "A commons for a supply chain in the post-COVID-19 era," *The Milbank Quarterly*, vol. 98, no. 4, pp. 1058–1090, 2020.

[93] S. M. Disney and D. R. Towill, "On the bullwhip and inventory variance produced by an ordering policy," *Omega*, vol. 31, no. 3, pp. 157–167, 2003.

[94] T. H. Davenport, "Competing on analytics," *Harvard Business Review*, vol. 84, no. 1, pp. 98–107, 2006.

[95] T. Tambunan, "SME development, economic growth, and government intervention in a developing country," *Journal of International Entrepreneurship*, vol. 6, no. 4, pp. 147–167, 2008.

[96] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[97] J. Wei *et al.*, "Chain-of-thought prompting elicits reasoning in large language models," in *NeurIPS*, vol. 35, 2022, pp. 24824–24837.

[98] K. E. McKone-Sweet, P. Hamilton, and S. B. Willis, "The ailing healthcare supply chain: A prescription for change," *Journal of Supply Chain Management*, vol. 41, no. 1, pp. 4–17, 2005.

[99] J. Webster and R. T. Watson, "Analyzing the past to prepare for the future: Writing a literature review," *MIS Quarterly*, vol. 26, no. 2, pp. xiii–xxiii, 2002.

[100] G. E. P. Box, G. M. Jenkins, G. C. Reinsel, and G. M. Ljung, *Time Series Analysis: Forecasting and Control*, 5th ed. Hoboken, NJ: John Wiley & Sons, 2015.

[101] D. Silver *et al.*, "Mastering the game of Go with deep neural networks and tree search," *Nature*, vol. 529, no. 7587, pp. 484–489, 2016.

[102] E. M. Rogers, *Diffusion of Innovations*, 5th ed. New York, NY: Free Press, 2003.

[103] M. E. Porter and M. R. Kramer, "Strategy and society: The link between competitive advantage and corporate social responsibility," *Harvard Business Review*, vol. 84, no. 12, pp. 78–92, 2006.